Fabrication and Characterization of Nanoscale NiO Resistance Change Memory (RRAM) Cells With Confined Conduction Paths

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***Abstract*—This paper introduces a novel structure for filamentary-conduction-type resistance change memory. The new structure utilizes the fringing field from the top metal electrode to effectively confine the position of the filaments in the cell. Nanoscale (100–50 nm) memory cells are fabricated using nickel oxide (NiO) thin film as the resistance-switching layer to demonstrate the feasibility and functionality of the new structure. The fabricated nanoscale memory cells achieved low reset current (100–200 μA), narrower distribution in low-resistance states, and higher on/off ratio than those of the standard structure. In addition, good scalability down to 50 nm was demonstrated.**

***Index Terms*—Metal-oxide memory, nanoscale memory cell, nickel oxide (NiO) thin film, resistance change memory (RRAM, ReRAM), Resistive memory, RRAM characterization.**

I. Introduction

METAL-OXIDE resistance change memory (RRAM) is one of the strong candidates for a future nonvolatile memory due to its good scalability, fast switching speed, and superb compatibility to the CMOS process [1]–[4]. Although the switching mechanism is not yet fully understood, a filamentary-conduction model is well established for some of the metal-oxide memories, such as NiO and TiO_2 [5]–[8]. The filamentary-conduction model proposes that resistance switching originates not from the entire electrode area but from a small localized area ("filaments") under the metal electrode. Recent studies have also proposed that the position of the filaments changes as the memory cell is programmed and erased, which results in a wide distribution of programmed states for the filamentary-conduction-type memories [9], [10]. Such wide resistance distribution, which not only reduces the minimum on/off ratio but also increases the failure rate of the cell, is one of the critical issues that must be improved for the filamentary-conduction-type memory.

Motivated by the goal to improve the resistance distribution, a novel memory cell structure for **filamentary-conduction-type memory** is proposed and demonstrated in this paper. **The novel structure confines the possible locations of the filaments to an area comparable to a single filament** ($\sim$10 nm) by significantly reducing the effective programming area, which was previously limited to the physical size of the patterned electrodes. The effectively confined filaments in this new structure result in a narrower distribution of the low-resistance states (LRSs). This work demonstrates the feasibility and functionality of this structure by presenting the characteristics of the nanoscale memory cells (100–50 nm) using NiO film, which includes dc I–V characteristics, switching speed, and distribution of resistance values.

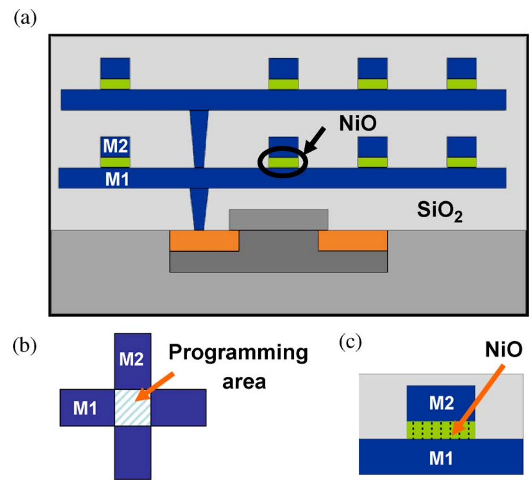

Fig. 1. (a) Simplified schematic of the conventional structure for RRAM. The resistance-switching material is sandwiched between (M2) the top electrode and (M1) the bottom electrode. (b) In conventional structures, the programming area is determined by the size of the electrodes. In this case, the programming area is given as M1 width $\times$ M2 width. (c) The electric field is uniformly distributed inside the resistance-switching material in the case of parallel-plate structure.

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II. Benefits From the Novel Structure

A. *Confinement of the Conduction Path in an RRAM Cell*

Most metal-oxide RRAM cell structures are composed of vertically stacked metal/metal-oxide/metal layers resembling the conventional parallel-plate capacitor [1], [3], [4] (Fig. 1). However, the conventional structure is not favorable for the filamentary-conduction-type memories because the two parallel-plate electrodes uniformly distribute the electric field across the entire area of a memory cell. To confine the formation

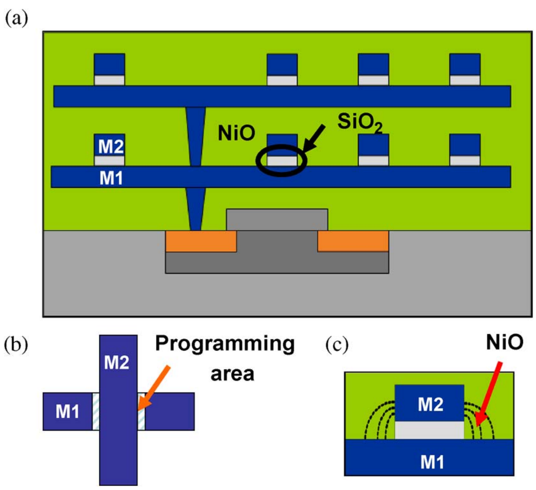

Fig. 2. (a) Simplified schematic of the proposed structure for filamentary-conduction-type RRAM. The resistance-switching material connects the side wall of (M2) the top electrode to (M1) the bottom electrode. (b) The programming area in the proposed structure is independent of the width of M2 since the conduction comes from the side wall. (c) In the proposed structure, the fringing field is utilized, and therefore, the conduction is more confined.

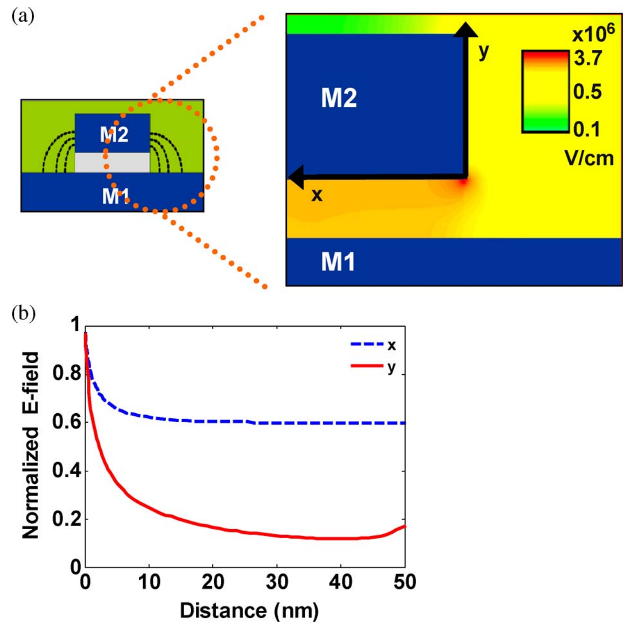

Fig. 3. (a) Simulation result showing the strength of the electric field around the M1 and M2 when 1 V is applied between the two electrodes. (b) Along the x-coordinate, which corresponds to the programming region in the parallel-plate structure (conventional), the distribution of the electric field is relatively uniform. In contrast, along the y-coordinate, which corresponds to the programming region in the proposed structure, the electric field rapidly decreases. Therefore, the simulation result shows that the conduction is more confined in the proposed structure [12].

of the filaments more effectively, we propose a structure where the resistance-switching material is laterally placed at the sides of the top metal electrode. In this structure, the resistance-switching material connects the side wall of the top electrode to the bottom electrode, while the top and bottom electrodes are isolated by an insulating layer at the cross-point (Fig. 2). As changing the location of the resistance-switching material, the distribution of electric field in the memory cell is significantly altered. In the conventional structure, the electric field inside the resistance-switching material is relatively uniform (x-coordinate in Fig. 3). Thus, the position of the filaments could easily vary as the cell is programmed and erased, assuming that the probability of having a filament is proportional to the strength of the electric field. On the other hand, in the proposed structure, the electric field rapidly decreases in the positive vertical direction along the side wall of the top electrode due to the asymmetry of the structure (y-coordinate in Fig. 3). Therefore, the formation of the filaments can be restricted to a very small area near the bottom corner of the top electrode, and a tighter distribution of the programmed states is expected from the proposed structure compared to the conventional structure. Furthermore, by utilizing the fringing field, an effective programming area smaller than the size of the electrode can be achieved beyond the lithographic limit. In this respect, the proposed structure is a good solution to achieve a small programming area until the lithography technology allows for patterning of the size comparable to that of the filament (< 10 nm). The location of the filament would simply be determined by the area of the electrode when the cell size approaches the physical size of the filament in the future.

B. Protection of the Programming Region by Self-Encapsulation

The encapsulation of the metal-oxide memory cells is crucial in RRAM structures since the chemical composition of the metal oxides is very sensitive to the ambient oxygen. The switching characteristics could degrade if the programming regions have been exposed to any contamination including the air during the fabrication or characterization steps [11]. The programming regions in the proposed structure are well protected because not only the programming regions are deep inside the metal-oxide film but also both top and bottom electrodes are formed prior to the metal-oxide deposition (Figs. 4 and 5). On the other hand, in the conventional structure, encapsulation by an additional dielectric is required, and the stack of metal-oxide/metal electrodes must be deposited *in situ* to protect the interfaces from contamination [11]. Therefore, the proposed structure is less sensitive to the process conditions, resulting in more consistent characteristics.

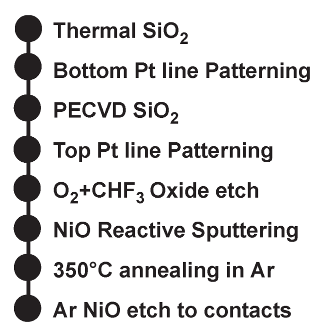

Fig. 4. Key process steps for the fabrication of the proposed structure [12].

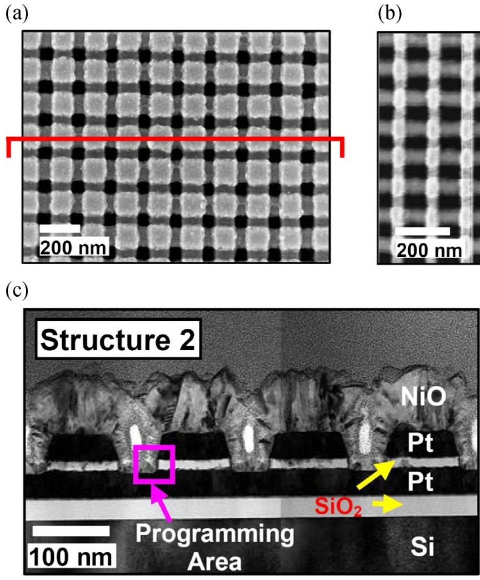

Fig. 5. Top-view SEM and cross-sectional TEM of the fabricated memory cells using the proposed structure (Structure 2). (a) 100 nm × 100 nm size memory cells. (b) 50 nm × 50 nm size memory cells. (c) Cross section of the structure along the red cutline in (a), after [12].

III. Fabrication of the NiO Memory Cells Using the Proposed Structure

The fabrication steps for the proposed structure are as follows (Fig. 4). First, 500-nm-thick silicon dioxide is thermally grown on the Si substrate for electrical isolation. Then, the bottom electrode metal lines (Pt, 50 nm in thickness) ranging in width from 100 to 50 nm are patterned on the substrate using ebeam lithography. A thin silicon dioxide layer (17 nm) ("intermetal oxide") is then deposited over the metal lines by PECVD at 350 °C. The top electrode metal lines (Pt) of the same size as the bottom electrode are then patterned crossing the bottom metal lines using ebeam lithography. The parts of the intermetal-oxide layer exposed between the top metal lines are etched by $CHF_3 + O_2$ plasma to open the contacts to the bottom metal lines. No mask is required in this etching step due to the high etch selectivity between SiO_2 and Pt. The 100-nm-thick NiO film is deposited on the top of the fabricated cross-point structure using reactive sputtering under 4% O_2 partial pressure at 200 °C, followed by an anneal at 350 °C for 30 min. Finally, the contacts to the top and bottom metal electrodes are made by etching NiO and patterning the metal pads.

Fig. 5 shows the top-view SEM of the fabricated memory cells and the cross-sectional TEM of the memory cells sharing a bottom metal line.

IV. Experimental Results and Discussion

The fabricated devices with the proposed structure (Fig. 5, **Structure 2**) were electrically characterized for dc characteristics, switching speed, and programming cycles. In this section, some of the characteristics will be discussed by comparing them with the measured characteristics of the test structures in which the NiO film is inserted between two metal electrodes resembling the conventional structures (Fig. 6, **Structure 1**).

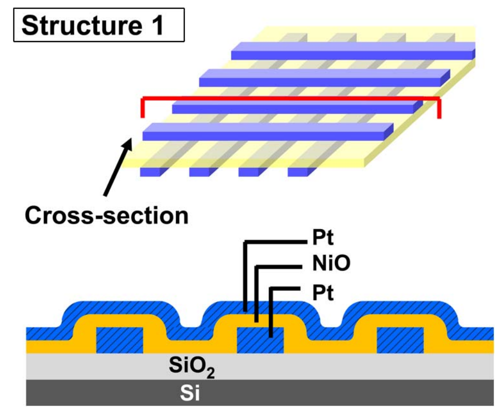

Fig. 6. Schematic of the test structure where NiO film is sandwiched by the two metal electrodes (Structure 1). This test structure was fabricated to resemble conventional structures and to be compared with the fabricated proposed structure.

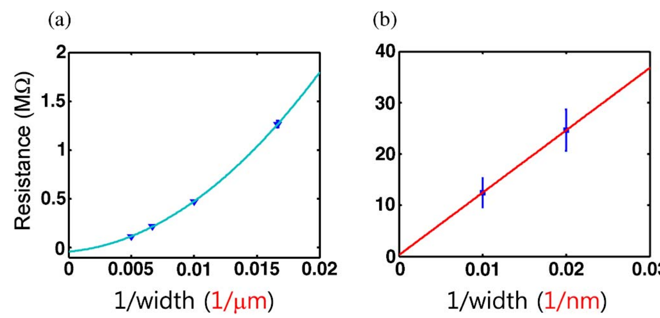

Fig. 7. Resistance of the as-fabricated fresh memory cells as a function of the size of the electrodes. (a) Measurement from micrometer-scale capacitorlike memory cells using square-shape electrodes. The conduction is a quadratic function of the width of the electrodes. (b) Measurement from the memory cells with the proposed structure (the bottom metal width is equal to the top metal width). The conduction is proportional to the width of the bottom metal line in the proposed structure.

A. DC Characteristics of the Proposed Structure

One of the most noticeable characteristics of the proposed structure is shown in Fig. 7. The measured conduction from the fresh cells (before any switching) is **not proportional to the area** of the cells (top metal line width × bottom metal line width) but **proportional to the width of the bottom metal lines** [Fig. 7(b)]. This property is one of the unique properties that distinguish the proposed structure from the conventional capacitorlike structure since the conduction of the as-fabricated fresh cells with the conventional structure is proportional to the area of the cells [Fig. 7(a)]. This result shows that the current mainly flows through the side wall of the top electrode, and therefore, the conduction is nearly independent of the width of the top electrode. The experimental result (Fig. 7) thus supports the idea that the proposed structure utilizes the fringing field from the top electrode during the electrical conduction.

Fig. 9 shows the dc I–V characteristics of the memory cells of the proposed structure, measured by dc voltage sweeps. First, the new memory cells in the fresh state require a forming process with a forming voltage of 3–4 V to have the subsequent SET (high to low resistance switching) and RESET (low to high resistance switching) transitions. The SET current is limited by an off-chip transistor at 90 μA (Fig. 8), and the minimum

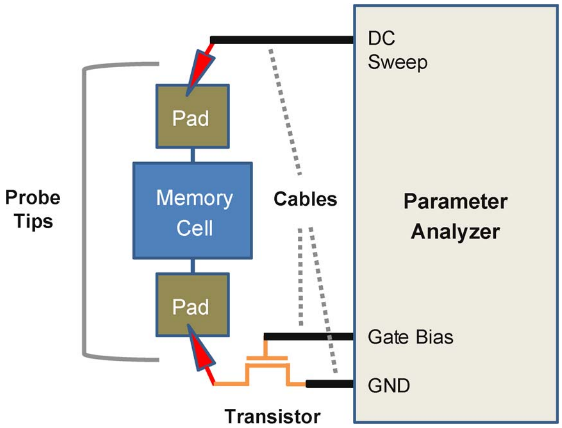

Fig. 8. Schematic of the measurement setup for dc characterization. An off-chip transistor is soldered onto a probe tip in order to suppress the overshoot current during the measurement. This setup still introduces some overshoot current due to the parasitic capacitance from the metal pads and the probe tips.

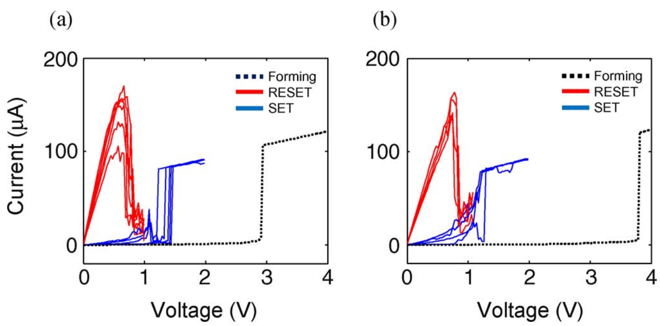

Fig. 9. DC I–V characteristics from the memory cells with the proposed structure. Multiple RESET and SET processes are shown with the solid lines. The forming processes are shown with the dotted lines. The SET current was limited at 90 μA by an off-chip transistor. (a) DC I–V from a 100-nm cell. (b) DC I–V from a 50-nm cell.

RESET current is about 160 μA for the 50-nm cell [Fig. 9(b)]. In the measurement setup, the off-chip transistor is directly soldered onto a probe tip as shown in Fig. 8. We observe that the nanoscale metal electrodes are destroyed due to large overshoot current during the SET transition if the off-chip transistor is not used for current limiting. It is also demonstrated that the key dc I–V characteristics of the proposed structure are similar to those of typical NiO RRAM cells with conventional structures [1], [3], [7].

Fig. 9(b) proves the functionality of the memory cells down to 50 nm by 50 nm cell size, verifying good scalability of the proposed structure.

B. Switching Speed of the Memory Cells

Switching speed of the new memory cells was measured by applying voltage pulses and measuring the current from a series resistor connected to the memory cell (50-Ω internal resistance of the oscilloscope was used). Fig. 10 shows that the low resistance to high resistance switching (RESET) takes about 180 ns when the RESET current is approximately 400 μA ($\sim$72 pJ of programming energy for RESET). In contrast, the high resistance to low resistance switching (SET) could not be accurately measured from the same measurement setup because

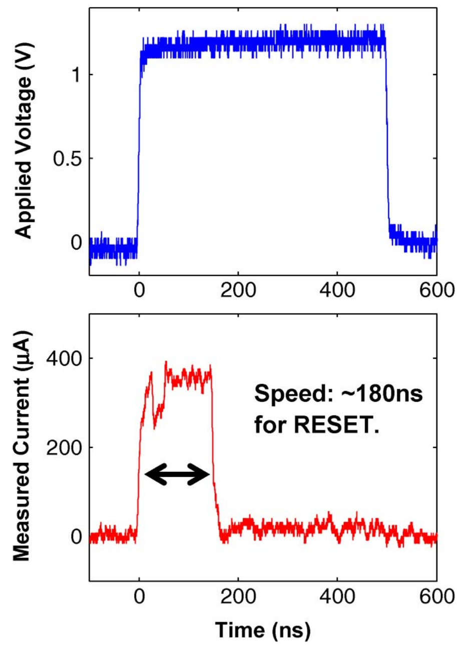

Fig. 10. Pulse test results for a RESET process from the 100-nm cell with the proposed structure. (a) Applied voltage pulse. (b) Measured cell current. The device switched from low resistance to high resistance in 180 ns. The RESET current is about 400 μA, and the energy consumed is about 72 pJ.

the switching speed is too fast to be observed for the parasitic of the measurement setup. Based on the fact that the device was switched by 10-ns pulses, we estimate that the SET process is shorter than 10 ns.

C. Resistance-Switching Cycle Test

Fig. 11 shows the read resistance values of the memory cells over 2000 dc programming cycles. The SET current was limited by the off-chip transistors connected to the memory cells during the tests. The memory cells from the proposed structure do not show any degradation over the 2000 cycles [Fig. 11(b)]. In addition, the minimum on/off ratio is about 10 for the 100-nm cell and is about 15 for the 50-nm cell [Fig. 11(b)].

The distributions of the read resistance values are collected based on the results from the programming cycle test. The distribution results (Fig. 12) demonstrate that the proposed structure (Structure 2) shows a narrower distribution of LRS compared to Structure 1 of which NiO is simply sandwiched by top and bottom electrodes. It should be noted that our measurement data are limited to the case where the overshoot current is relatively large during the SET, influenced by the parasitic capacitance from the off-chip transistor and the device pads [13]. The narrower LRS distribution in Structure 2 is probably a direct consequence of the smaller effective programming area as similarly observed by Baek *et al.* [10].

On the other hand, Structure 2 shows a wider distribution of high-resistance states (HRSs) compared to Structure 1 (Fig. 13). The wider HRS distribution is a possible issue when the programming area is aggressively reduced in filamentary-conduction-type RRAM cells since the ruptured filaments contribute larger portion to the total off-state leakage as the conduction area scales down.

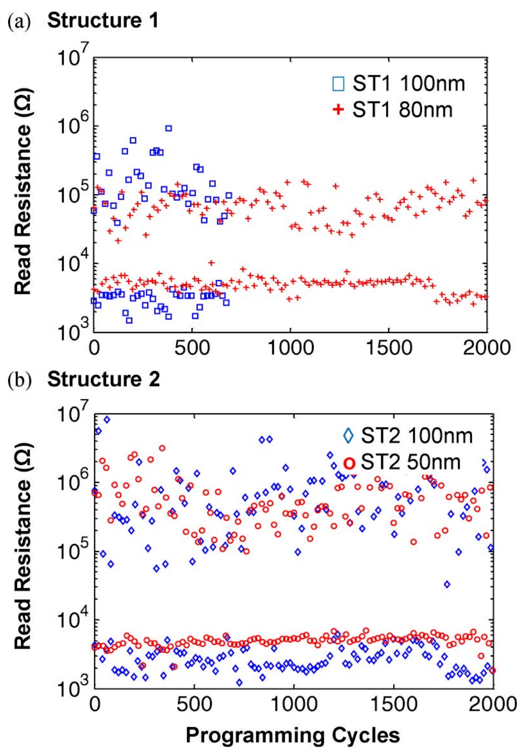

Fig. 11. DC programming cycle test from the Structure 1 and the Structure 2. The SET current was limited to about 100 μA by off-chip transistors during the tests. (a) The 100-nm cell with the Structure 1 failed at about 700 cycles. (b) No degradation or failure was observed from the 100-nm cell and 50-nm cell with the proposed structure (Structure 2) over 2000 cycles.

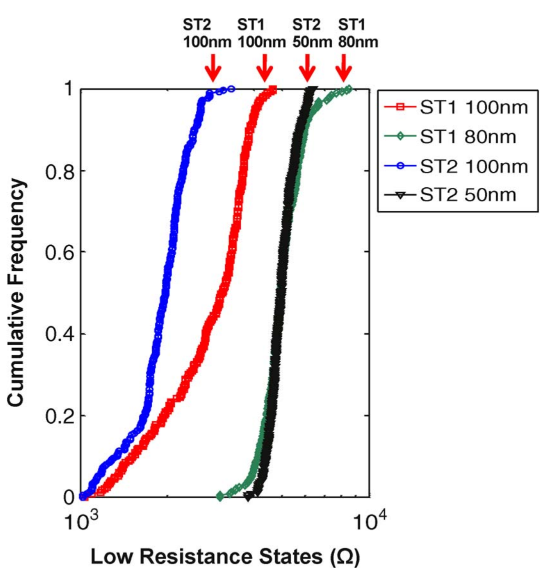

Fig. 12. Distribution of the LRSs from four different structures: From the left, 100-nm cell with Structure 2 (ST2), 100-nm cell with Structure 1 (ST1), 50-nm cell with Structure 2, and 80-nm cell with Structure 1. This graph shows that Structure 2 has tighter distribution of the LRSs compared to Structure 1.

V. Conclusion

A novel structure for a filamentary-conduction-type RRAM has been proposed and successfully fabricated to demonstrate enhanced characteristics. In this structure, extremely small effective programming area is achieved using fringing field from the top electrode. Specifically, the process steps and characterization of the NiO RRAM memory cells using this structure were experimentally demonstrated: The memory cells showed a minimum RESET current of 160 μA, and the scalability is demonstrated down to 50 nm by 50 nm cell size. From the pulse measurement, the switching speed ($\sim$180 ns) and the programming energy ($\sim$72 pJ) of the RESET transition were measured. Finally, the memory cells were characterized over 2000 programming cycles, which demonstrates that the proposed structure (Structure 2) yields a tighter distribution of LRSs compared to that of the parallel-plate structure (Structure 1).

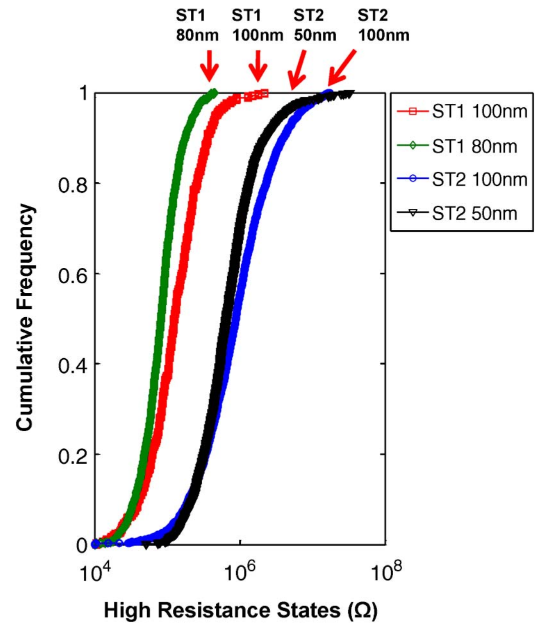

Fig. 13. Distribution of the HRSs from four different structures. This graph shows that Structure 2 shows wider distribution of HRSs compared to Structure 1.

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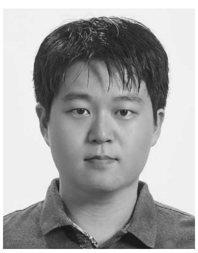

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